Supplement: cross-disciplinary reframings of the Goldbach–Chen frontier

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This supplement records, in compact form, the cross-disciplinary reframings of the main paper, *A Finite-Range Half-Power Artifact at the Goldbach–Chen Frontier*, together with the empirical probes moved out of the body. Each reframing recasts the *same* singular-series governance in a different external language; we keep them for the individual vocabularies and the two honest negatives, not as independent results. "The main text" refers to that paper, and every number is reproducible from the `run_*.py` drivers in the repository.

1 Cross-disciplinary reframings, in brief

Each item below is one fact—$\mathfrak{S}(N)$ sets the local densities—seen in a different mirror; none is an independent finding, so we state each in a sentence.

Honest negatives. Two recastings return *negative* results, which we record because they bound what the structure does *not* buy. *Persistent topology:* for the binary problem the additive-complexity sublevel filtration is a disjoint matching, so its 0-persistence is just the energy spectrum and all higher homology vanishes—genuine higher-order structure appears only at the ternary level, which is anti-clustered. *Market resilience:* reading N as a market cleared by two prime inputs, the Chen semiprime substitute is *fair-weather liquidity*—under a supply shock the substitute spread decays linearly, $S_C(\phi) = S_C(0)(1-\phi) \to 0$, with a negligible resilience premium.

The remaining vocabularies, one line each.

- **Arithmetic ground-state energy.** With $H(a,b) = \Omega(a) + \Omega(b)$, Goldbach is "$\min H = 2$"; the melting temperature $\beta^\star(N)$ at which order localises is an almost deterministic function of the singular series, $\mathrm{corr}(\beta^\star, \log \mathfrak{S}) = -0.92$.
- **Adversarial robustness.** The Goldbach graph is random-robust but structure-fragile: deleting 95% of primes at random breaks no N in the window, yet deleting one reduced class mod 3 (half the primes) destroys 59%.
- **Control / value of information.** A residue-aware search ordering cuts the expected per-N Goldbach-search cost from 5.5 to 2.5 tries at budget 13, the gain per prime tracking $(\ell-1)/(\ell-2)$.
- **Information content.** Conditioning $\log R_1$ on N mod 30030 leaves a residual of 0.11% that is white noise: up to that modulus the comet is almost entirely a singular-series codeword.

- **Discrete geometry of $p + rs = N$.** The Chen solutions of a typical N concentrate against the Goldbach edge—median Li–Liu exponent $a \approx 1.30$, far below the 1.9 threshold—the geometric form of the residual-exponent finding.
- **Infinite formulation / column generation.** The selection MIP of the main text has an infinite-dimensional reading whose columns are representation certificates priced by reduced cost; the content is selection under structure, not feasibility.

2 Further empirical probes (moved from the main text)

For length and focus, the following experiments were moved out of the main text. Each is reproducible from its `run_*.py` driver in the repository and, like every reframing here, returns to the same singular series $\mathfrak{S}(N)$.

2.1 Minimum-variation selection and the Chen-rescue threshold

These two diagnostics use the selection language of the previous section. The minimum-variation branch is a diagnostic of the representation cloud, not an invariant of Goldbach–Chen arithmetic; the endpoint branch $s_N \approx 1$ reflects the objective's preference for small p.

Smoothness is a Pareto trade-off. Minimizing the total spatial variation $\sum |s_N - s_{N-2}|$ of the selected location $s_N = q_N/N$ yields a geometrically very smooth branch (total variation 0.0040 over 201 consecutive even integers) that nonetheless *switches type in* 28% *of its steps* and uses semiprimes 72% of the time (Fig. 1). Forbidding semiprimes (the pure Goldbach branch, zero type switches) raises the variation to 0.0246, a factor 6.2 rougher. One cannot simultaneously maximize geometric and type smoothness.

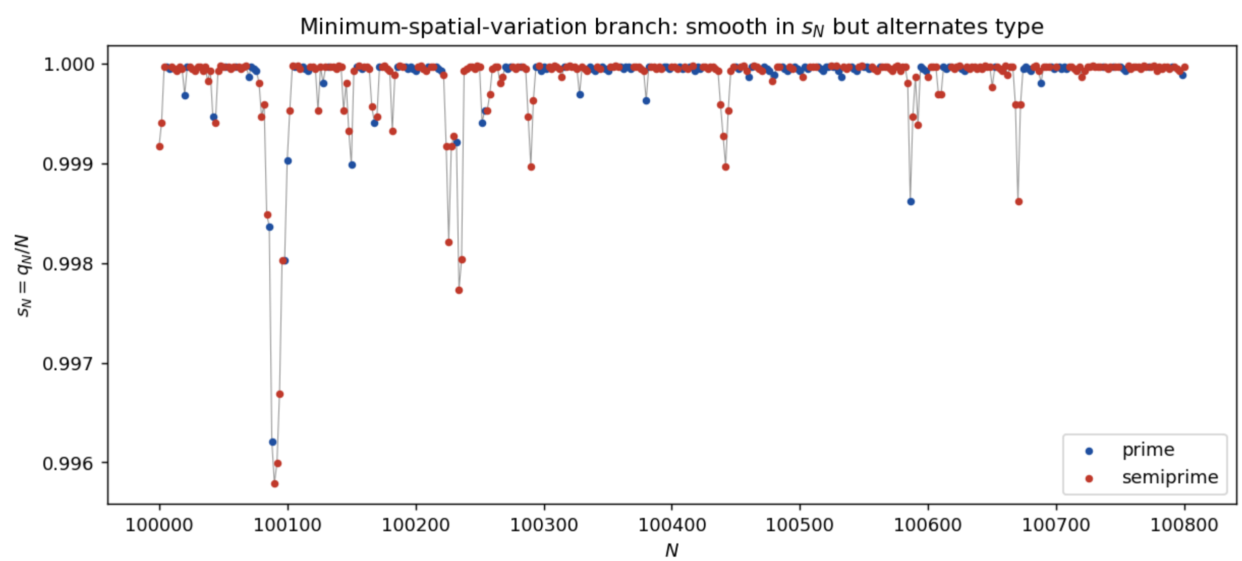

Figure 1: Minimum-variation branch (Gurobi). The optimum hugs $s_N \approx 1$ and is smooth in s_N, yet its type (blue prime / red semiprime) alternates.

A unimodal Chen-rescue threshold. Restricting representations to a balanced band $p \in [N/2 - k,\, N/2]$ and measuring the fraction of N that *require* a semiprime (no prime pair in the band) gives a unimodal curve in k (Fig. 2). For $k \gtrsim 0.005\,N$ the prime channel essentially never fails; the rescue fraction peaks near 38–40% at $k \sim 20$–40; for $k = 1, 2$ it falls again because often no representation exists at all. The peak shifts right with N (roughly as $\log^2 N$), where the expected number of balanced prime pairs $\sim 2C_2\,\mathfrak{S}(N)\,k/\log^2(N/2)$ drops to $O(1)$. This is the regime captured by the residual exponent $a_{\mathrm{Res}}(N)$ of the main text: Chen's flexibility becomes *necessary* precisely when near-perfect balance is imposed.

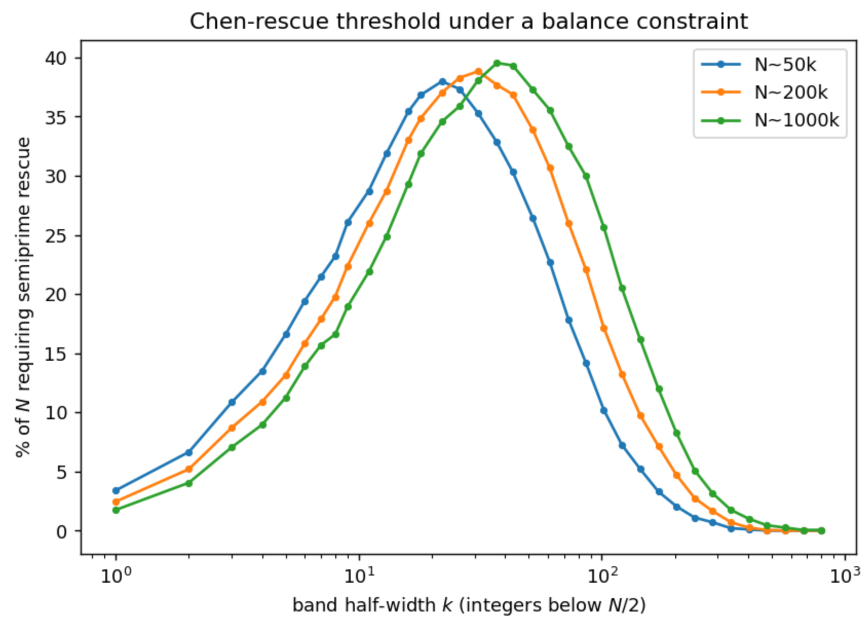

Figure 2: Fraction of N that require a semiprime rescue versus band half-width k, for three scales of N. The curve is unimodal with peak scaling like $\log^2 N$.

2.2 Weak continuity of the location measures

The pointwise measure μ_N of q/N is noisy, but averaging over a window $N \in [X, X + H]$ produces a stable density (Fig. 3). At $X = 10^6$: *(i)* splitting a window into two same-scale subsamples, the total-variation distance between their densities falls monotonically from $2.1 \cdot 10^{-3}$ to $3 \cdot 10^{-4}$ as H grows from 125 to 4000 (a concrete form of weak convergence); *(ii)* the prime-q density is within total variation 0.028 of uniform on $(0, 1)$, with a mild endpoint upturn (the $1/(\log(xN)\log((1 - x)N))$ effect); *(iii)* prime and semiprime locations agree to total variation 0.019, the semiprime location slightly skewed toward larger q (mean $q/N = 0.513$ versus 0.500 for primes). The main paper's analytic appendix gives the weighted Hardy–Littlewood explanation: after normalization both the inherited singular series and the Mertens channel sum cancel, leaving the uniform law for q/N in either channel (conditional on the weighted channel asymptotics).

2.3 Optimal transport: the reflection defect

Under the reflection $T(x) = 1 - x$ (which sends p/N to q/N), compare the prime location law α (density of p/N) with the reflected layer laws $T_\#\beta_k$. The reflection defects in W_1 are $\Delta_1 =$

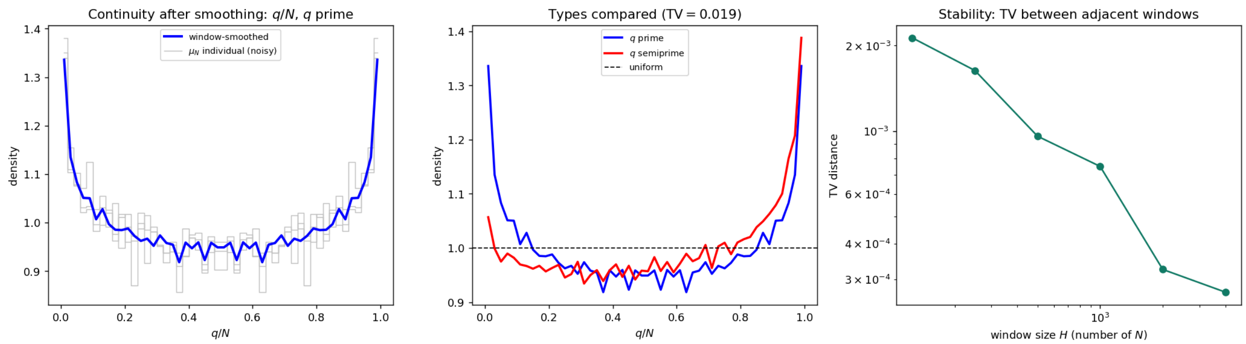

Figure 3: Weak continuity of q/N. Left: noisy pointwise μ_N versus the window-smoothed density. Center: prime versus semiprime location densities, both near uniform. Right: same-scale sampling distance versus window size H (log–log).

$W_1(\alpha, T_\# \beta_1) = 0.040$ and $\Delta_2 = 0.032$, and the optimal Chen mixture attains $\Delta_{\leq 2} = 0.032$ at $\lambda^\star = 0$: the reflected *semiprime* cloud matches the primes *better* than the reflected primes do, a 22% "transport collapse" (stable in X), driven by unbalanced semiprimes with a small factor (Fig. 4).

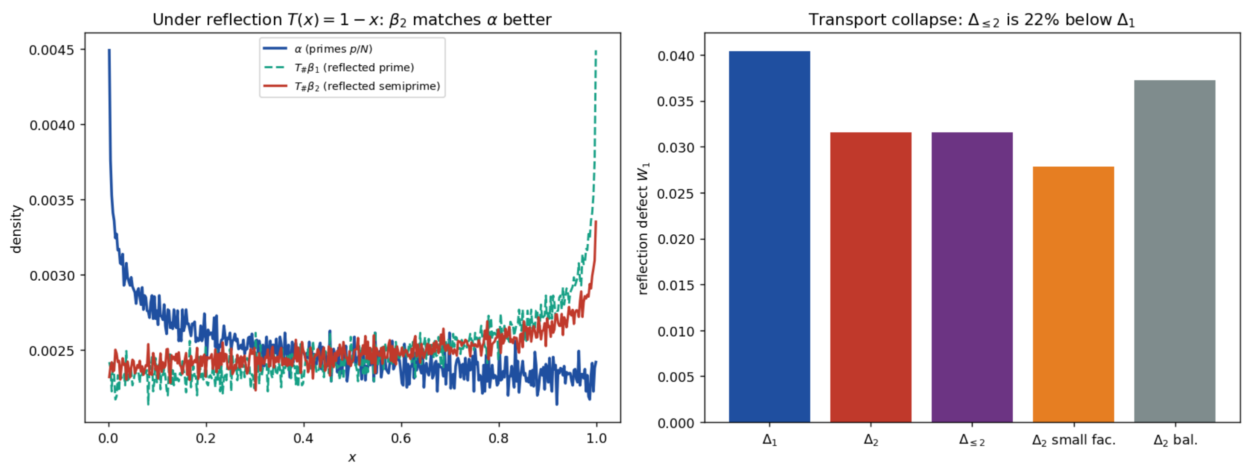

Figure 4: Reflection transport. Left: prime law α and the reflected prime and semiprime laws. Right: admitting semiprimes lowers the reflection defect by 22%.

2.4 Topology of valleys

Normalising each count by its own singular factor ($R_1/\mathfrak{S}$, $R_{\leq 2}/(\mathfrak{S} + \mathfrak{S}^{1/2} W)$) and computing the 0-dimensional sublevel persistence of the detrended sequences, the valleys of the Goldbach and Chen comets largely *coincide*: the anomalies correlate at $+0.64$, and at the 200 most fragile N the Chen anomaly (0.29) is as deep as the Goldbach one (0.30). Chen does *not* topologically fill the Goldbach valleys (only the deepest 30% are softened, Fig. 5); the rescue is one of magnitude, not of valley structure, because both channels share the constraint that p be prime. The correlation $+0.64$ is a within-window descriptive statistic over a strongly arithmetically dependent sequence (consecutive even N), not an i.i.d. estimate.

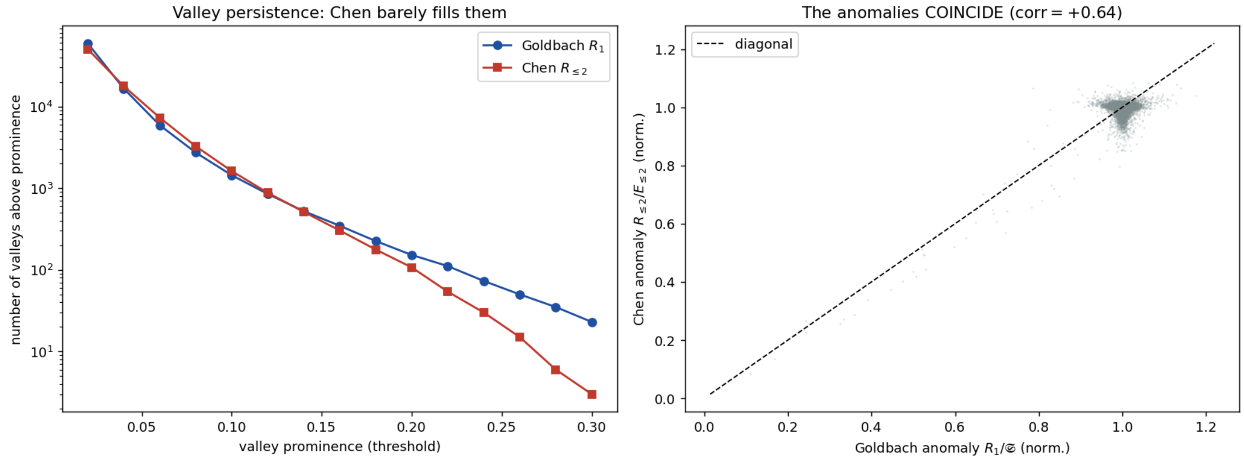

Figure 5: Valley persistence. Left: Chen has nearly as many valleys as Goldbach. Right: the normalised anomalies coincide (+0.64).

2.5 Statistical dynamics of the frontier

The prime share is almost a deterministic function of the singular series: $\theta(N)$ is 94% explained by $\mathfrak{S}(N)$ alone (adding $N \bmod 30$ contributes nothing further), and the increment $\Delta\theta_N$ is 98% explained by the jump $\mathfrak{S}(N) \to \mathfrak{S}(N+2)$. The residual is small (25% of the spread of θ) and is *not* white noise (lag-1 autocorrelation -0.21 after detrending): the frontier carries little irreducible randomness, being governed by the local arithmetic of N (Fig. 6). These percentages are descriptive R^2 values over an arithmetically dependent window, not i.i.d. fits.

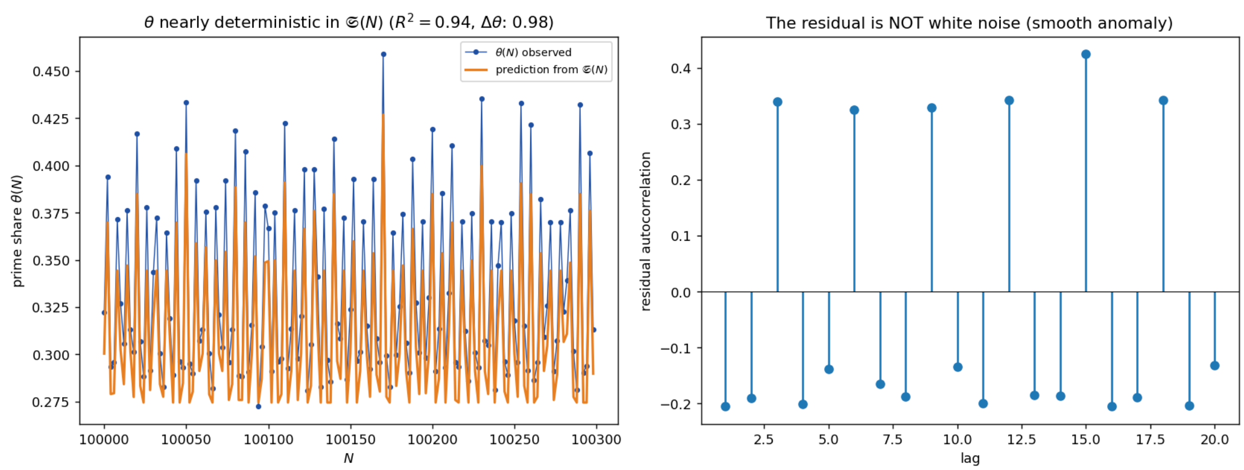

Figure 6: Dynamics. Left: $\theta(N)$ tracks its singular-series prediction. Right: the residual autocorrelation, small and non-white.